PROJECT DOCUMENTATION

Interactive 3D Flipbook Product Catalogue

A responsive, brandable page-flip catalogue website built with React, TypeScript, Vite and Tailwind CSS



Catalogue cover image used as page 1 of the flipbook (sample placeholder artwork)

Package name	flipbook-product-catalogue (version 1.0.0, private)
Type	Static front-end web app (no backend, API keys or database)
Stack	React 19, TypeScript, Vite, Tailwind CSS 4, Motion, Lucide icons
Catalogue size	10 pages, 16 images, 21 source components
Developer credit in app	Developed by Kazi Jawad (configurable)

Contents

Contents 2

1. Project Overview 3

2. Feature Overview 4

3. Technology Stack 5

4. Architecture 6

5. The Page-Flip Engine 7

6. The 10-Page Demo Catalogue 8

7. Ready-Made Page Templates 13

8. Controls and Interface Reference 14

9. Customisation Guide 15

10. Build and Deployment 16

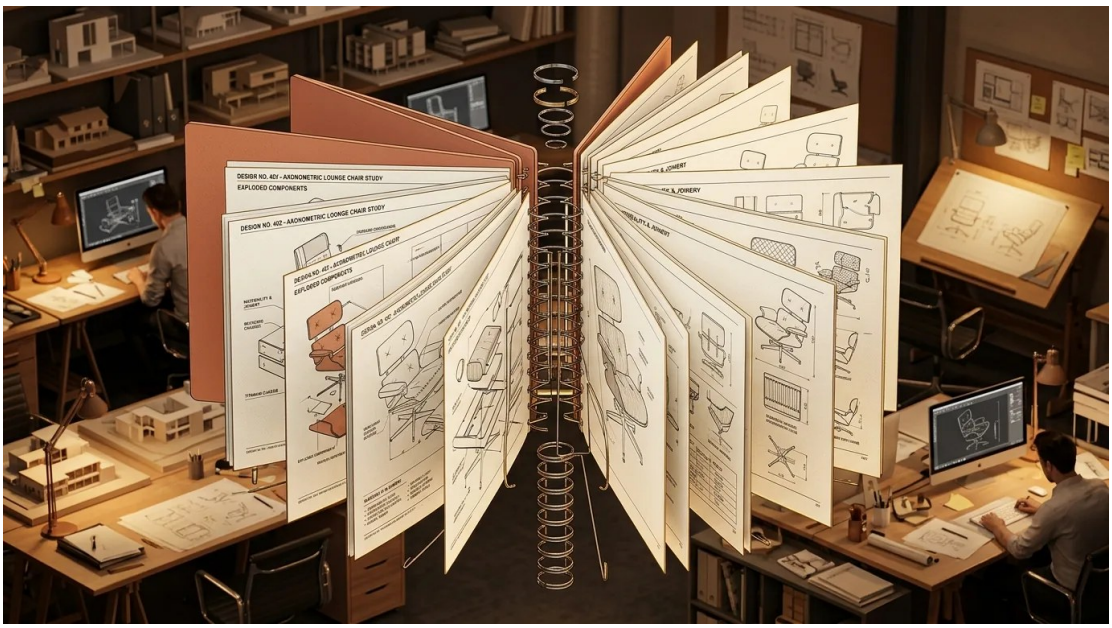
11. Review Notes and Recommendations 17

> **How this document was produced.** It is based on a read-through of the project source (components, configuration, README and the Fiverr gig file) and its image assets. The project was **not installed, built or run** for this document, so build size and performance figures are quoted from the project README rather than measured.

1. Project Overview

This project is an **interactive 3D flipbook product catalogue**: a website that presents a catalogue as a bound book whose pages physically turn in 3D when a visitor clicks, swipes, drags or uses the keyboard. It is designed as a **white-label template**. A developer replaces the brand details and page images, runs one build command, and publishes the resulting static files on any web host.

The bundled demo is a furniture catalogue (office seating, lounge pods, tables, storage, materials and a BIM toolkit). The furniture content is placeholder material for demonstration; the codebase itself is industry-neutral and is positioned for furniture, real estate, fashion, construction, manufacturing, restaurants, lookbooks and portfolios.



Concept render used on the "Flipbook Concept" page (page 2)

At a glance

Aspect	Detail
Purpose	Turn a static PDF-style catalogue into an engaging, shareable, mobile-friendly web experience
Delivery model	Static files in `dist/` after `npm run build`; deployable to Netlify, Vercel, Cloudflare Pages, GitHub Pages or cPanel hosting
Customisation	Two config files (`brand.ts`, `pages.ts`) plus an images folder
Page types	Plain full-page image, or a custom React component per page
Interaction	3D page turn, drag-to-peel, zoom, fullscreen, autoplay, keyboard, index, thumbnails, share, enquiry, right-click menu, help
Source volume	About 4,776 lines of TypeScript, TSX and CSS across the `src` folder
Runtime requirement	Node.js 18 or newer for development and build; any static host for deployment

2. Feature Overview

Experience features

- **Realistic 3D page turn** - each leaf bends as it swings over a spiral binding, with shadows, driven frame-by-frame by `requestAnimationFrame`.
- **Drag-to-peel** - on touch and mouse, dragging a page follows the pointer; releasing past roughly 28% of the drag range completes the turn, otherwise the page springs back.
- **Synthesised page-flip sound** - generated live with the Web Audio API (filtered pink noise); no audio files are shipped, and it can be muted.
- **Zoom** - toggles a zoomed view of the current page (drag-to-turn is disabled while zoomed).
- **Autoplay slideshow** - advances every 4.5 seconds and loops back to page 1 after the last page.
- **Fullscreen mode** - uses the browser Fullscreen API and stays in sync if the user exits with Esc.
- **Fully responsive** - desktop, tablet and phone layouts; the control bar collapses secondary actions into a "More Options" overflow on small screens.

Navigation and discovery

- **Control bar** - table of contents, all-pages thumbnails, first / previous / next / last, and a "jump to page" number field.
- **Index modal** - a searchable table of contents; the search matches section titles, index titles and "page N".
- **Pages modal** - a responsive thumbnail grid (one to three columns) that updates automatically when pages are added.
- **Custom right-click menu** - replaces the browser menu with catalogue actions (previous, next, cover, back cover, index, pages grid, share, plus view toggles).

Engagement and sharing

- **Share modal** - copy link, native device share sheet (falls back to copy when unsupported), and email.
- **Concierge (enquiry) modal** - a contact panel showing phone, email, address and opening hours plus an enquiry form (name, email, project scope, message).
- **Closing page** - a branded thank-you page with contact links (email, LinkedIn, GitHub, website) configured in one place.

3. Technology Stack

Technology	Version (package.json)	Role in the project
React	^19.0.1	UI components and state (hooks only; no external state library)
TypeScript	^7.0.2	Typed components and configuration; `npm run lint` runs `tsc --noEmit`
Vite	^8.3.0	Dev server (port 3000) and production build
@vitejs/plugin-react	^6.1.1	React support in Vite
Tailwind CSS + @tailwindcss/vite	^4.3.3	Utility-first styling; brand accent exposed as CSS variables
Motion	^12.23.24	Animation library available to the UI
lucide-react	^0.546.0	Icon set for controls and modals
Google Fonts	loaded in index.html	Playfair Display (headings) and Plus Jakarta Sans (UI); needs internet access, otherwise system fonts are used

npm scripts

Command	What it does
`npm run dev`	Starts Vite on port 3000, reachable on the local network (`--host`)
`npm run build`	Builds optimised static files into `dist/`
`npm run preview`	Serves the production build on port 4173 for testing
`npm run lint`	Type-checks the project with the TypeScript compiler

Project structure

```
flipbook-product-catalogue/
  index.html            page shell, fonts, social-preview meta tags
  package.json          dependencies and scripts
  README.md             quick start, customisation and deployment guide
  FIVERR_GIG.md         ready-to-paste Fiverr listing, packages, FAQ
  vite.config.ts        React + Tailwind plugins, "@" path alias
  src/
    main.tsx            entry: applies brand colours, title, meta description
    App.tsx             owns all state; assembles header, stage, controls, modals
    index.css           Tailwind import, fonts, 3D helper classes, scrollbars
    config/
      brand.ts          name, tagline, colours, contact, closing links, credit
      pages.ts          the ordered list of catalogue pages
    components/         21 components (stage, controls, modals, page templates)
    utils/audio.ts      Web Audio page-turn sound
    assets/images/      18 .webp images
```

4. Architecture

The application is a single-page React app with one state owner. `App.tsx` holds the current page number and every view toggle, and passes values and callbacks down to presentational components. There is no router, store or network layer.

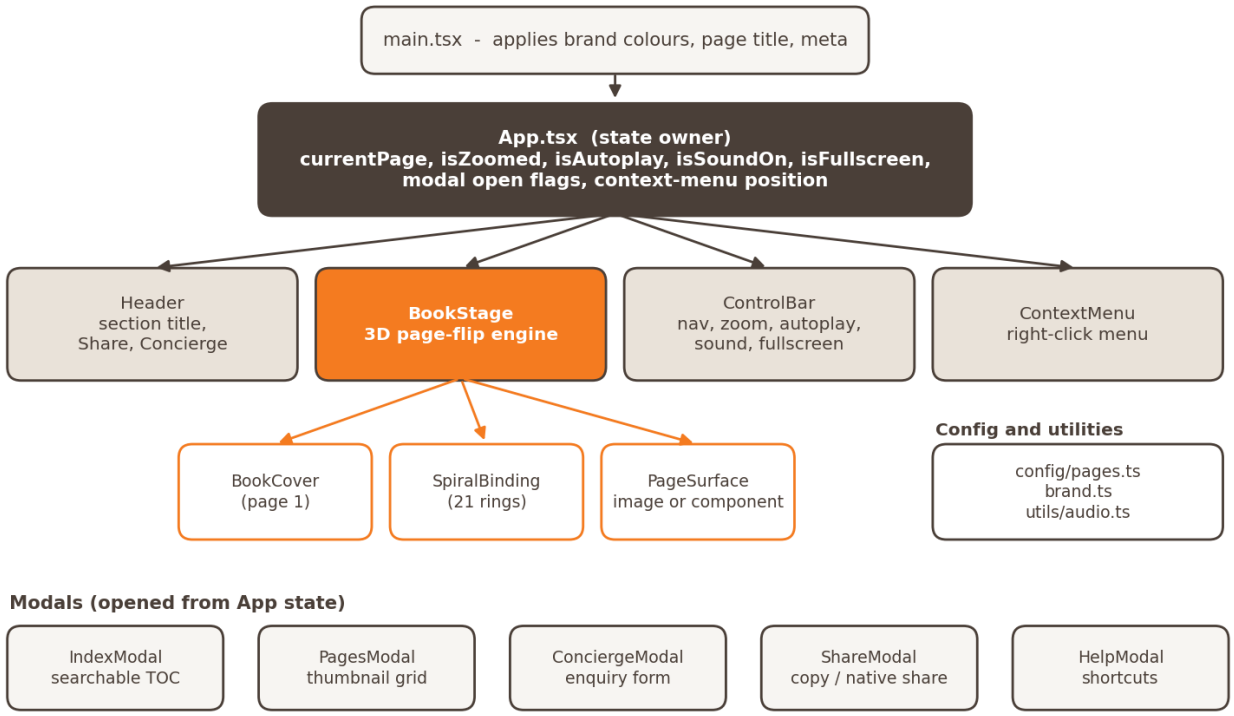


Figure 1 - Component architecture

State held in App.tsx

State	Purpose
currentPage	1-based page number; clamped between 1 and the total page count
isZoomed / isAutoplay / isSoundOn / isFullscreen	View toggles, shared by the control bar and the right-click menu
isIndexOpen, isPagesOpen, isConciergeOpen, isShareOpen, isHelpOpen	One flag per modal
contextMenu {isOpen, x, y}	Position of the custom right-click menu

Because every navigation source (buttons, keyboard, thumbnails, index, autoplay, right-click menu) writes to the same `currentPage` state, the page-flip engine reacts to one value only, which keeps behaviour consistent across all of them.

5. The Page-Flip Engine

The 3D effect lives in `BookStage.tsx` (about 480 lines) with help from `SpiralBinding.tsx`. Rather than relying on a third-party flipbook library, it implements its own animation loop.

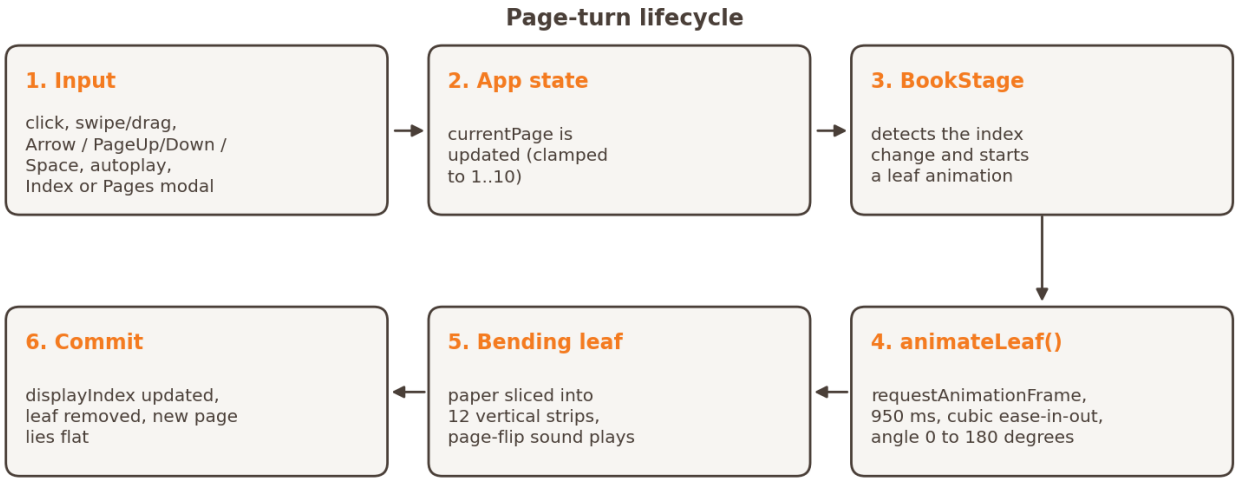


Figure 2 - What happens when the page changes

Key mechanics

Mechanic	Implementation
Bending paper	The turning leaf is sliced into 12 vertical strips (`STRIPS = 12`) so each strip can rotate slightly differently, producing a curved sheet
Animation	Custom `requestAnimationFrame` loop; a full turn lasts 950 ms (`TURN_MS`) using a cubic ease-in-out curve
Leaf angle	0 degrees = page flat on the right; 180 degrees = fully turned over the spiral to the left
Perspective	CSS perspective of 2600 px on the stage, with preserved 3D transforms and hidden back faces
Drag-to-peel	Pointer events with pointer capture; an 8 px dead zone, progress measured over 80% of the page width, commit when progress exceeds 0.28
Performance	Page faces are memoised (`PageSurface`) so the 60 fps animation never re-renders page content
Binding	`SpiralBinding` draws 21 rings by default (configurable via a `count` prop)
Keyboard	Right Arrow, Page Down or Space = next; Left Arrow or Page Up = previous; ignored while typing in inputs
Sound	`utils/audio.ts` synthesises ~0.28 s of band-pass filtered pink noise (sweeping about 1400 to 750 Hz) per turn

Design note. Pages are rendered by `PageSurface`, which picks what to draw in this order: the cover component for page 1, then a custom React component if the page defines one, then a plain image. This is why a page can be either a flat picture or a fully interactive layout.

6. The 10-Page Demo Catalogue

The page list is defined in `src/config/pages.ts`. Each entry has an id, page number, header label (`sectionTitle`), index label (`indexTitle`), an image, an optional custom component and a description used as alt text.

#	Section title	Rendered as	Image shown
1	About us and introduction	BookCover component (cover image)	01_cover-page
2	The flipbook concept	FlipbookConceptPage component	02_flipbook-concept-render
3	Collection directory and index	FurnitureTableOfContentsPage component	03_collection-directory
4	Executive seating and ergonomics	Full-page image	04_page-modern-office-table
5	Lounge and acoustic pods	Full-page image	05_page-director-table
6	Desking and architectural tables	Full-page image	06_page-conference-table-ravishing
7	Storage and partitions	Full-page image	07_page-conference-table
8	Materials and sustainability	Full-page image	08_page-curated-bookshelf
9	Architect and BIM toolkit	Full-page image	09_page-bookshelf-range
10	Contact and thank you	BackCoverShowroomsPage component	10_back-cover

Page images 4 to 9 are six designed product-page layouts (office table, director table, conference tables and bookshelves). Page-turn images below are exactly the files used by the app.

Page artwork used in the flipbook



Page 4 image - "Modern Office Table" layout



Page 5 image - "Director Table" layout



Page 6 image - "Ravishing Conference Table" layout

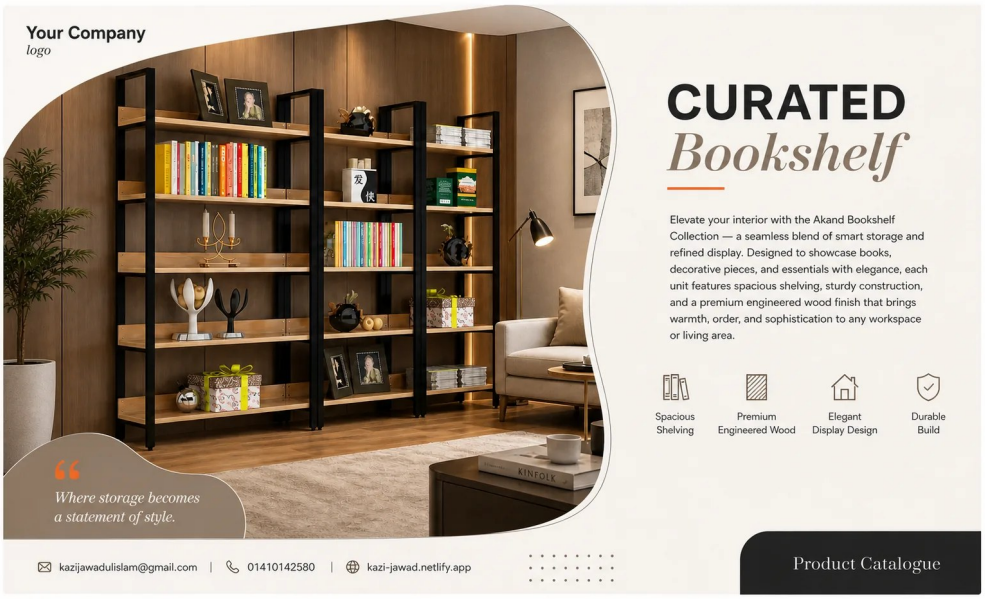


Page 7 image - "Conference Table" layout

Page artwork (continued)



Page 3 image - collection directory



Page 8 image - "Curated Bookshelf" layout



Page 9 image - "Bookshelf" range layout

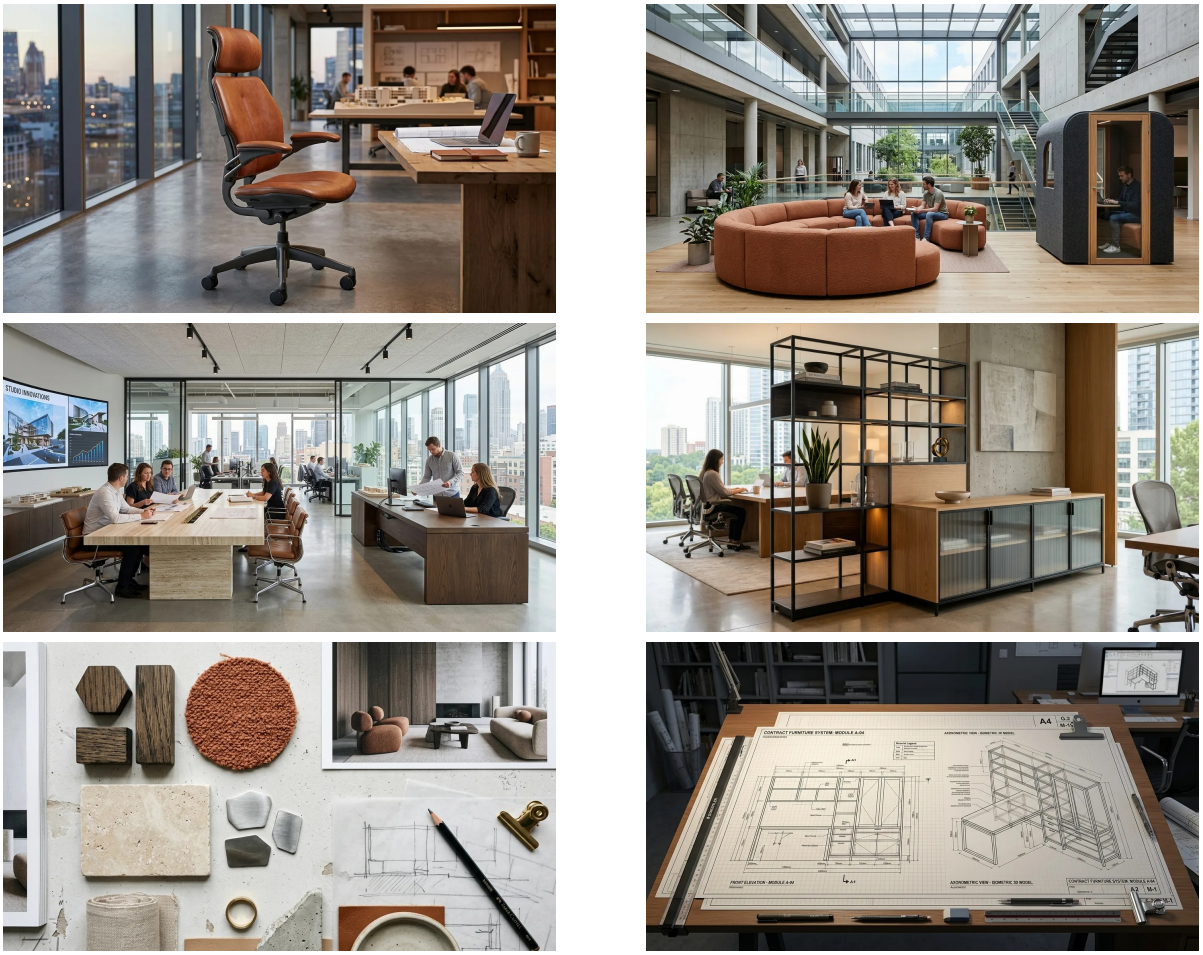


Page 10 image - back cover

7. Ready-Made Page Templates

Six additional page components are included in `src/components`, each paired with its own image. They show how a page can be a designed React layout instead of a flat picture, and serve as templates (the README points to `ExecutiveSeatingPage.tsx` as the model to copy).

Component	Image	Theme
ExecutiveSeatingPage	11_component-executive-seating	Task and executive chairs, finishes picker
LoungeAcousticPage	12_component-lounge-acoustic	Modular lounge seating and acoustic pods
DeskingTablesPage	13_component-desking-tables	Boardroom tables and height-adjustable desking
StoragePartitionsPage	14_component-storage-partitions	Credenzas and room partitions
MaterialsSustainabilityPage	15_component-materials-swatches	Materials, finishes and sustainability
ArchitectBimToolkitPage	16_component-bim-toolkit	CAD / BIM download resources



Images used by the six page templates: executive seating, lounge and acoustic pods, desking and tables, storage and partitions, materials swatches, and the BIM toolkit.

> **Note.** In the current `pages.ts`, pages 4 to 9 use flat images, so these six components are included in the source but are not wired into the page list. To use one, import it in `pages.ts` and set `component: ExecutiveSeatingPage` on the matching entry.

8. Controls and Interface Reference

Layout

The screen is a three-part column: a **header** (current section title, a "live view" pill, Share and Concierge buttons), the central **book stage**, and a bottom **control bar**. Everything else opens as a modal on top.

Ways to turn pages

Method	How
Mouse / touch	Click the page edge, or drag the page and release
Keyboard - next	Right Arrow, Page Down or Space
Keyboard - previous	Left Arrow or Page Up
Control bar	First, previous, next, last buttons and a jump-to-page number field
Index modal	Click any entry in the searchable table of contents
Pages modal	Click any thumbnail
Right-click menu	Previous, next, cover page, back cover
Autoplay	Automatic every 4.5 s, loops at the end

Control bar buttons

Table of contents, all-pages thumbnails, first page, previous, jump-to-page, next, last page, zoom (or reset zoom), start/pause slideshow, mute/enable page-flip audio, enter/exit fullscreen, catalogue shortcuts and help, and "More Options" (mobile overflow).

Modals

Modal	Contents
Index	Searchable list of pages; highlights the current page
Pages	Thumbnail grid of every page; auto-populates from pages.ts
Concierge	Contact details (phone, email, address, hours) and an enquiry form with name, email, project scope (Commercial Office Space, Executive Boardroom, Hospitality and Lounge, Bespoke Architectural Project) and message
Share	Copy link, native share sheet, email
Help	Keyboard and gesture guide (next, previous, 3D flip, zoom, fullscreen)

9. Customisation Guide

1. Brand and contact details - src/config/brand.ts

This single file white-labels the catalogue. The accent colour is applied across the whole UI through CSS variables set in `main.tsx`.

Field	What it controls	Current value in the project
name, shortName	Brand name in the UI, page title and modals	"Your Company"
tagline, subtitle	Marketing line and the "Product Catalogue" label	"Durability. Smart Design. Innovation." / "Product Catalogue"
siteDescription	Meta description	Interactive flipbook description
colors	Accent, hover, dark UI, surfaces, text	Accent #F47B20, hover #E06D16, dark #4A3F38
contact	Phone, email, address, hours shown in the Concierge modal	Demo phone and address; the email is the developer's own
closing	Badge, heading, message and link list for the final page	Email, LinkedIn, GitHub and website links
credit	Small "Developed by" footer line; `show: false` hides it	"Developed by Kazi Jawad", shown

2. Pages - src/config/pages.ts

```
{
  id: 4,
  pageNumber: 4,
  sectionTitle: "EXECUTIVE SEATING & ERGONOMICS",  // header label
  indexTitle:   "03. Executive Seating",           // index / thumbnails
  image:        page4Image,                        // full-page image
  description:  "Short description (alt text)",
}
```

- **Image-only page**: import the image and set `image`. Recommended 3:4 portrait, about 1500 x 2000 px, .webp or .jpg under 200 KB.
- **Custom page**: add `component: YourPage`.
- **Add or remove pages** by adding or removing entries; page count, index, thumbnails and navigation update automatically.

3. Images and social preview

Drop artwork into `src/assets/images/` and update the imports at the top of `pages.ts`. For link previews, edit the Open Graph and Twitter tags in `index.html`.

10. Build and Deployment

Run locally

```
npm install
npm run dev        # http://localhost:3000
npm run build      # outputs to /dist
npm run preview    # test the production build (port 4173)
```

Deploy

Host	How
Netlify / Vercel / Cloudflare Pages	Connect the repository; build command `npm run build`; output directory `dist`
cPanel / shared hosting	Upload the contents of `dist/` to `public_html`
GitHub Pages	Add `base: '/repo-name/'` in `vite.config.ts`, then publish `dist/`

Performance (as stated in the README)

The README states the catalogue loads fast, with about 2 MB of images and about 93 kB of gzipped JavaScript. These numbers were not re-measured for this document.

Commercial packaging (FIVERR_GIG.md)

The project ships with a complete Fiverr gig listing: title, description, tags, packages, extras, FAQ, buyer requirements, a gallery plan, a per-order delivery checklist and message templates. The suggested packages are:

	Basic	Standard	Premium
Name	Starter Flipbook	Branded Catalogue	Full Custom Catalogue
Pages	Up to 10	Up to 20	Up to 40
Delivery	3 days	5 days	10 days
Revisions	1	3	Unlimited
Deployment to client host	No	Yes	Yes
Working enquiry form	No	Yes	Yes
Custom page layouts and domain setup	No	No	Yes
Suggested price	$50	$120	$280

Suggested extras range from $15 (PDF to optimised page images) to $40 (second-language version).

11. Review Notes and Recommendations

These points come from reading the source. None stops the project working; they are worth checking before delivering it to a client.

Finding	Detail	Suggested action
Share link page number is not used	The Share modal builds a link ending in `?page=N`, but no code reads that parameter, so opening a shared link always starts at page 1	Read `page` from the URL in `App.tsx` to set the initial `currentPage`
Enquiry form does not send data	The form shows a confirmation and closes, but submits nowhere (the README says this too)	Connect to Formspree, EmailJS or your own endpoint in `ConciergeModal.tsx`
Some demo text is hard-coded	`BookCover.tsx` contains the text "FORMA" and the showroom cities, and the page components contain furniture copy, so changing `brand.ts` alone will not rebrand them	Search the components for demo text before white-labelling
Page descriptions do not match images	Descriptions in `pages.ts` for pages 4 to 9 mention seating, pods and BIM, while the images show tables and bookshelves	Update descriptions (they act as alt text) to match the artwork
Personal details in brand.ts	The default contact email and the closing-page links are the developer's own, while the company name is a placeholder	Replace with the client's details for every delivery
Six page components unused	Executive seating, lounge, desking, storage, materials and BIM components exist but are not referenced by `pages.ts`	Wire them in, or remove them to keep the delivery lean
Static page title and preview text	`index.html` uses generic "Product Catalogue" text for the title and social tags (the title is overridden at runtime, social tags are not)	Set Open Graph text per client
node_modules shipped in the zip	The uploaded archive includes `node_modules` (about 150 MB unpacked, with a Windows-specific native binary)	Exclude it from deliveries, as the Fiverr checklist already advises
Fonts need internet	Google Fonts are loaded from the web; offline visitors see fallback fonts	Self-host fonts if offline use matters

Summary. The project is a clean, dependency-light static React template with a custom 3D page-flip engine, a rich set of viewing controls and a ready-made sales package. With the small fixes above, it is ready to be re-branded and delivered as a catalogue product.

Appendix - image files used in this document

All 16 artwork images (01 to 16) are the actual image assets referenced by the project source, converted from .webp to .jpg for portability. Images 17 and 18 are diagrams created for this document. All are supplied in the accompanying zip.